

# Part 4

## The Model Editor and the vehicle metadata

The Model Editor is TF3's Blueprint Editor and asset previewer in one. This part walks through importing the FBX set, filling every metadata struct a rail vehicle needs, validating, making icons, and the rail-specific extras: fake bogies, multiple units and buy-menu groups.

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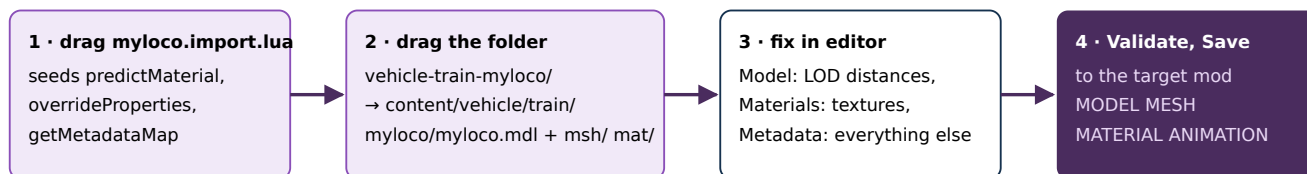
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Everything in this part is from the Model Editor, Vehicle Basics, Vehicle Types, Advanced Topics and Repaint Mods pages unless marked.

# 1 · Set up and first start

- 1 Run `ModelEditor.bat` (Windows), `ModelEditor.sh` (Linux) or `ModelEditor.command` (macOS) from the game's installation folder. Two windows open: a console and the UI. [manual](#)
- 2 On first start it writes a config file under `%APPDATA%/Transport Fever 3/` (or `~/.local/share/Transport Fever 3`, or `~/Library/Application Support/Transport Fever 3/`) and asks for the **userDataPath**. Find it in game: Settings ► Advanced ► Open user data folder. Use forward slashes only. [manual](#)
- 3 Restart. The open dialog now browses the base game and every mod in the **staging area**. [manual](#)
- 4 Options ► Im-/Export: tick **Use Blender Coordinate System**; decide the Overwrite... switches (meshes/animations/materials/textures) for re-imports. [manual](#)
- 5 Options ► Config: set **Enable Auto Update** to *On Focus* if your paint program writes textures slowly (black textures otherwise). [manual](#)

## 2 · Import: template first, then the FBX folder



Saving has no overwrite confirmation; the save dropdown lets you deselect MODEL / MESH / MATERIAL / ANIMATION.

**Figure 4.1 · Import order** [manual](#). The `.import.lua` template is optional but is what lets you re-import the FBX a dozen times without re-typing metadata: its `getMetadataMap()` is applied “as is” to the models imported afterwards.

- Folder name decides the content path: `vehicle-train-myloco/` → `vehicle/train/myloco/myloco.mdl`; use `+` instead of the last `-` to put several models in one folder (multiple units). [manual](#)
- Drag a single FBX for one LOD, the folder for all LODs. [manual](#)
- Bulk: drag a `.txt` with `ModRefName;<modid>` and `Path;...` lines to import many models. [manual](#)
- After import, the **Model** window shows the node tree with mesh references and material/animation counts; hover a line to isolate the mesh; double-click a name to copy it for metadata. [manual](#)

## 3 · The metadata structs, field by field

| Struct              | Fields (rail vehicle) <span>manual</span>                                                                                                                                                                                                                          | From the TSC side                                                                            |
|---------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------|
| transportVehicle    | carrier "RAIL" · transportModes {TRAIN, ELECTRIC_TRAIN} · engineTransportModes {ELECTRIC_TRAIN} (locos) · compartmentsList · loadSpeed · comfortFactor · arrivalDelay / departureDelay ms · reversible · filterTags {"default"} · groupFileName · multipleUnitOnly | Electric subsystem → ELECTRIC_TRAIN; diesel/steam → TRAIN only. Delays from door animations. |
| landVehicle         | engines { {power kW, tractiveEffort kN, type} } · topSpeed m/s · weightEmpty kg · weightMaxPayload kg · friction · brakeDeceleration                                                                                                                               | Sim<br>MaxPower/MaxForce/MaxSpeed, bp Mass (TF3 guide Part 1 §6 conversions).                |
| railVehicle         | config.axles {node names} · config.fakeBogies {one list per LOD}                                                                                                                                                                                                   | Axle names from the name map; usually { }, { }.                                              |
| seatProvider        | drivingLicense "RAIL" · crewModels · seats { {animation, crew, forward, group, transf} }                                                                                                                                                                           | Driver empty becomes the crew seat on import.                                                |
| soundConfig         | soundSet.name (path to .snd relative to the .mdl) · effects { event = file }                                                                                                                                                                                       | Pick a base-game set <span>check</span> paths.                                               |
| emissions           | noise / pollution: score 0-100 or -1 (auto), or explicit idle/speed/power                                                                                                                                                                                          | Balance against power.                                                                       |
| availability        | yearFrom, yearTo (0 = open end)                                                                                                                                                                                                                                    | Prototype years.                                                                             |
| description         | name, description as _("KEY") → strings.json                                                                                                                                                                                                                       | Engine name.                                                                                 |
| cost                | price (-1 for auto <span>check</span> )                                                                                                                                                                                                                            | -                                                                                            |
| colorConfig         | configs { { {r,g,b} }, ... } for colour-blend materials                                                                                                                                                                                                            | Liveries.                                                                                    |
| transformatorConfig | reference to a .trf.lua + params                                                                                                                                                                                                                                   | Only for custom logic (Part 5).                                                              |

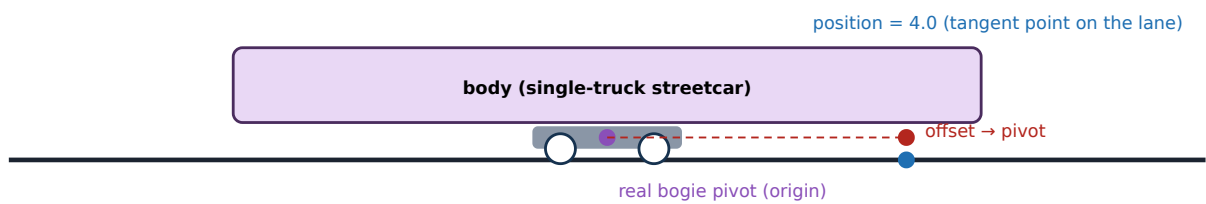
The Metadata window's + button adds structs; the gear offers templates; Shift-click an arrow to unfold a whole property. manual

## 4 · Compartments and seats

```
compartmentsList = {
  { loadConfigs = {
    { cargoEntry = { capacity = 80,
      cargoTypeSet = { cargoClassesIncluded = { "PASSENGERS" }, cargoClassesExcluded = {},
        cargoTypesIncluded = {}, cargoTypesExcluded = {} },
      loadIndicator = "", seats = {} },
    toHide = {} },
  } },
},
```

- One compartment is usually enough; several load configs when the same space can carry different cargo splits (coal only, or coal + ore). manual
- A locomotive normally has an empty `compartmentsList = {}` and a crew seat; a coach has passenger capacity and passenger seats; a wagon has a cargo class and, for visible loads, a `loadIndicator` with a `cargoBay` on a node. manual
- Seats: place `|seat_sitting` empties in Blender for passengers (the import brings them in), or add them in the editor with the **Seat Locators** render option on. manual

## 5 • Fake bogies



**Figure 4.2 • A fake bogie gives a one-bogie body its second alignment point** manual. `group` = node, `position` = distance along the lane from the root (positive = front), `offset` = shift along the tangent at that point, `upright` optional. Up to two per node; listed per LOD.

```
railVehicle = { config = { axles = { "w1", "w2" },
  fakeBogies = { { { group = "body", offset = 0, position = 4.0, upright = false } }, -- LOD 0
    { { group = "body", offset = 0, position = 4.0, upright = false } } } } --
LOD 1
```

Validation fails if the list length differs from the LOD count (“size (= 2) must match number of lods (= 3) or be empty”). manual

## 6 • Validation

The top-bar validate button checks metadata and reports one issue at a time; run it again after each fix. manual Messages you will meet on a rail vehicle:

| Message <span>manual</span>                                                                               | Meaning / fix                                                                           |
|-----------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------|
| <code>railVehicle/configs/fakeBogies:</code><br>size (= 2) must match number of<br>lods (= 3) or be empty | One fakeBogies list per LOD, even <code>{}</code> .                                     |
| Referenced mesh node 'XYZ' not<br>found (key:<br>'seatProvider.seats[10]')                                | Seat group names a node that does not exist (renamed? LOD?).                            |
| <code>transportVehicle/loadConfigs:</code><br>list of load configs must not be<br>empty                   | Every compartment needs at least one load config.                                       |
| Group 'body' has 2 materials but<br>mesh has 3 groups                                                     | Material count per node must equal the mesh's material groups; fix<br>slots in Blender. |

| Message <span>manual</span>                                     | Meaning / fix                                                                     |
|-----------------------------------------------------------------|-----------------------------------------------------------------------------------|
| Used technique ... requires missing mesh attribute ... UV1      | Material type needs a second UV set (AO) the mesh lacks; change type or add UV1.  |
| map_albedo has missing texture                                  | Texture path wrong or file not yet converted.                                     |
| Animation ... has keyframes starting after 0                    | Key frame 0.                                                                      |
| Model Group 'body' uses skinned mesh with an unskinned material | Vertex groups on a mesh that should be rigid: remove them or use a Skin material. |

## 7 • Icons and screenshots

Press **SCREENSHOT**: it generates the three UI icons ( `_store` , `_small_icon` , `_icon20` ) and moves them into the mod, plus orthographic and perspective renders in `screenshots/` under the user data path. The game needs `.tga` . Options ► Icons sets the stage track model, sizes, pixels per metre and which outputs to generate; Bulk Generate does a whole folder. manual

## 8 • Multiple units and buy-menu groups

```
-- myunit.mu.lua, next to the .mdl files
function data() return {
  vehicles = {
    { name = "myloco.mdl",    forward = true
  },
    { name = "mycoach.mdl",   forward = true
  },
    { name = "myloco.mdl",    forward = false
  }, -- flipped 180°
  },
  name = _("VEHICLE_MU_MYUNIT_NAME"),
  desc = _("VEHICLE_MU_MYUNIT_DESCRIPTION"),
  filterTags = { "default" },
  groupFileName = "",
} end
```

### Groups

Set `transportVehicle.groupFileName = "menu_myloco.mdl"` (relative path) on each variant to group liveries or lengths under one store entry. For a custom group icon, make a placeholder model with `multipleUnitOnly = true` and no `groupFileName` ; it is never sold, only shown as the parent. Icon style: up to four variants, 30 px apart, ≤ 327 px wide. manual

All parts must be within their availability years for the unit to appear. manual

## 9 • Repaints: the cheap second livery

A repaint mod changes only textures: new `.mtl` per re-textured material ( `map_albedo_opacity` , `map_normal` , `map_metal_gloss_ao` , `map_cblend_dirt_rust` ), a new `.mdl` per variant whose material references point at them in every LOD, new UI icons, new textures, a different modId and the original mod as dependency. manual This is how your TSC liveries become TF3 variants without touching the meshes.

## 10 · Field reference: the two sample vehicles as TF3 metadata

| Field                           | Sample Electric Engine                                                                                     | Example Steam Engine (SDJR 7F)                                                               |
|---------------------------------|------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------|
| landVehicle.engines             | { power = 6600, tractiveEffort = 300, type = "ELECTRIC" } (if TSC values are kW/kN <a href="#">check</a> ) | { power = 990, tractiveEffort = 157, type = "STEAM" } (1328 hp × 0.7457; 35.296 klb × 4.448) |
| landVehicle.topSpeed            | 64.8 (145 mph)                                                                                             | 26.8 (60 mph)                                                                                |
| landVehicle.weightEmpty         | 84000                                                                                                      | 68000 (+ tender as a separate vehicle)                                                       |
| railVehicle.config.axles        | { "w1","w2","w3","w4","w5","w6" }                                                                          | pony truck axle + 4 coupled axles; rods under drive                                          |
| transportVehicle.transportModes | { "TRAIN", "ELECTRIC_TRAIN" }; engine modes { "ELECTRIC_TRAIN" }                                           | { "TRAIN" }; engine modes { "TRAIN" }                                                        |
| transportVehicle.reversible     | true (double cab)                                                                                          | false                                                                                        |
| seatProvider                    | 2 crew seats, forward true/false, groups = the two headlight meshes                                        | 1 crew seat (driver), 1 crew seat idle (fireman)                                             |
| Emitters                        | none                                                                                                       | emitter_smoke_dark at the chimney; steam at cylinders optional                               |
| Parts events                    | headlights fwd/rev → front_forward / front_backward                                                        | lamps → front_forward_parts                                                                  |